Bharath S Hegde

me24btech11010@iith.ac.in | bharathshreepad@gmail.com | +91 8121116535 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.98
Narayana Junior College, Addagutta	Class XII (Telangana Board of Secondary Education)	2024	90.70%
Vikas the Concept School	Class X (Central Board of Secondary Education)	2022	92.00%

WORK EXPERIENCE

Engineering Intern | Schematic Engineering Industries - Pharmaceutical Equipment Manufacturer

- Performed CFD simulations using ANSYS Fluent and OpenFOAM to optimize heat exchanger geometry, improving heat transfer, and coolant flow characteristics
- Designed and optimized pressure vessel geometry using FEA to optimize optimize strength and withstand high pressure conditions, reduced stress concentrations, and minimize weight, cost.
- Performed thermodynamic analysis of a two-stage vapor compression refrigeration system to select compressors, condensers, vacuum pumps, and heat transfer fluids for pharmaceutical equipment

SKILLS

Programming Languages and Tools: Python (NumPy, OpenCV, matplotlib), C, MATLAB, LaTeX, SQL, Excel

Frameworks: ROS2, Linux (Ubuntu),

Softwares: FEA - Ansys, FreeCAD, SolidEdge(Professional); **CFD** - OpenFOAM, AnsysFluent; **CAD** - Fusion360, SolidEdge

PROJECTS

Truss Bridge Structure | (Intro to Mechanical Engineering Course Project)

- Designed and fabricated a truss bridge, experimentally evaluating structural behavior, stress, strain and failure modes under progressively increasing loads
- Performed finite element analysis (FEA) to estimate stress distribution and theoretical failure loads
- Developed an OpenCV-based image processing pipeline to track nodal displacement and estimate strain experimentally.

6-Bar Mechanism Dynamic Force Analysis | MATLAB (Kinematics and Dynamics of Mechanisms Course Project)

- Developed a MATLAB-based dynamic force solver for a planar six-bar mechanism using force and torque equations.
- Formulated and solved a 16 × 16 system of simultaneous equations to determine unknown joint reaction forces and driving torque throughout the mechanism cycle
- Generated plots of reaction forces, actuator torque, and dynamic loading as functions of crank angle for engineering analysis

Snake Bot - Modular Snake Robot | Sensors, Arduino, MATLAB, Fusion360, 3D Printing

- Developed a **control method** for **biologically inspired snake-like locomotion** using servo motors, implementing gait patterns across modular segments
- Integrated an **ultrasonic sensor for obstacle detection**, enabling the robot to halt or attempt path adjustment around detected obstacles
- Designed and fabricated a modular snake robot with changeable body segments to enable rapid prototyping and scalability

- Modeled the robot in Fusion 360 and optimized modular components for rapid fabrication using 3D printing

Tentacle Bot - Soft Robotics Manipulator/Gripper | *Python, Fusion360, 3D-Printing*

- Developed a **Python-based kinematic simulation** to predict manipulator deformation, workspace, and bending characteristics
- Implemented **closed-loop contact detection using servo current sensing** -- monitored real-time current draw to identify contact via current spikes
- Designed and fabricated a TPU-based soft robotic manipulator capable of multi-directional bending based on a research paper by Wang et. al
- Designed CAD models and assembled the prototype using TPU 3D-printed components
- Designed a tendon-driven 3-string actuation mechanism for controlled multi-directional bending

POSITIONS OF RESPONSIBILITY

Overall Head, Robotix - *Robotics Club*

- Organized member recruitment by creating robotics-focused technical assessments and conducting **15+ technical interviews**

Mechatronics Core , Tinkerer's Lab - *Mechatronics Club*

- Implemented **ROS2-based motion planning and manipulation** for a **6-DOF** robotic arm
- Designed CAD models of the gripper, robotic arm, and chassis for an autonomous WALL-E-inspired waste collection robot, optimizing components for manufacturability and assembly
- Contributed to the mechanical design and assembly of an autonomous delivery drone
- Conducted hands-on workshops on Arduino programming, robotics fundamentals, and Bluetooth-controlled RC vehicle for students

RELEVANT COURSES

Mechanical: Fabrication Lab, Digital Fabrication, Solid Mechanics, Manufacturing Science, Engineering Mechanics, Fundamentals of Robotics, Heat and Mass Transfer, Thermodynamics, Dynamics of Mechanisms

Mathematics: Linear Algebra, Calculus-1,2 , Differential Equations, Transform Techniques